

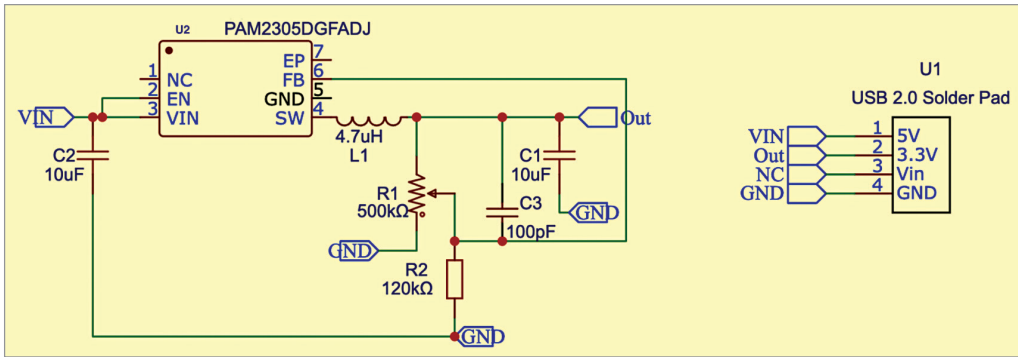


Fig. 6: Buck converter circuit for variable output voltage

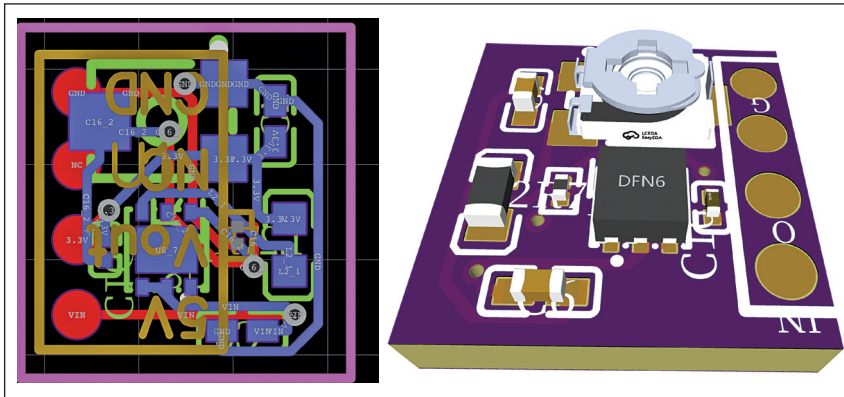


Fig. 7: Author's prototype

application circuit). The resistor values R1 and R2 can be selected from Table 2 to achieve the desired output voltage. For example:

- For 1.2V output: R1 = 100kΩ, R2 = 100kΩ
- For 1.5V: R1 = 150kΩ, R2 = 100kΩ
- For 1.8V: R1 = 200kΩ, R2 = 100kΩ
- For 2.5V: R1 = 380kΩ, R2 = 120kΩ
- For 3.3V: R1 = 540kΩ, R2 = 120kΩ

For variable voltage adjustment, R1 can be replaced with a 500kΩ to 600kΩ potentiometer while keeping R2 at 120kΩ. This allows for tuning the output voltage across a range (e.g., 1.2V to 3.3V) by adjusting the potentiometer wiper.

Circuit

According to the application circuit

in the datasheet, it includes a VIN pin for the input voltage. In the final circuit, SMD solder footprint points have been added for both input and output voltage connections. If integrating the converter design into another device, the circuit can be used directly.

There are two types of buck converters—variable and fixed. Both are built around the IC PAM2305DGFADJ and a few other components.

Fig. 4 shows the buck converter circuit configured for a fixed output voltage. In this fixed output configuration, resistors are connected to the feedback pin of the IC. SMD capacitors are used at both the input and output, along with an inductor connected to the SW pin for efficient voltage regulation. This setup requires a power supply of 3.2V to 3.3V. Resistor R1 is approximately 540kΩ (or 536kΩ, depending on

precision), and R2 is around 120kΩ.

Fig. 6 shows the buck converter circuit configured for a variable output voltage. In the variable output configuration, SMD capacitors are also used at both the input and output, along with an inductor

connected to the SW pin. The R1 resistor is replaced with a 500kΩ SMD potentiometer to allow output voltage adjustment. Fig. 5 shows the potentiometer, and Fig. 6 shows its placement within the circuit.

Construction and testing

The PCB layout of the variable voltage converter (see Fig. 6) can be designed using any PCB design software. After fabrication, the PAM-2305DGFADJ IC (DFN-6 package) should be carefully soldered onto the board. An inductor (4.7µH–10µH), input and output ceramic capacitors (typically 10µF), and the required resistors or potentiometer should be placed as per the data-sheet guidelines.

A power supply (up to 5.5V) should be connected, and the potentiometer can be adjusted to set the output voltage. For testing, a 5V USB input can be applied, and the output can be measured using a multimeter. The potentiometer may be rotated gradually to vary the output voltage (typically from 1V to 3.3V). Output stability and ripple can be observed using an oscilloscope. It is advisable to monitor temperature and check efficiency across different load conditions. Fig. 7 shows the final prototype assembled on the PCB. **EFY**

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